



National Oceanic and
Atmospheric Administration
U.S. Department of Commerce

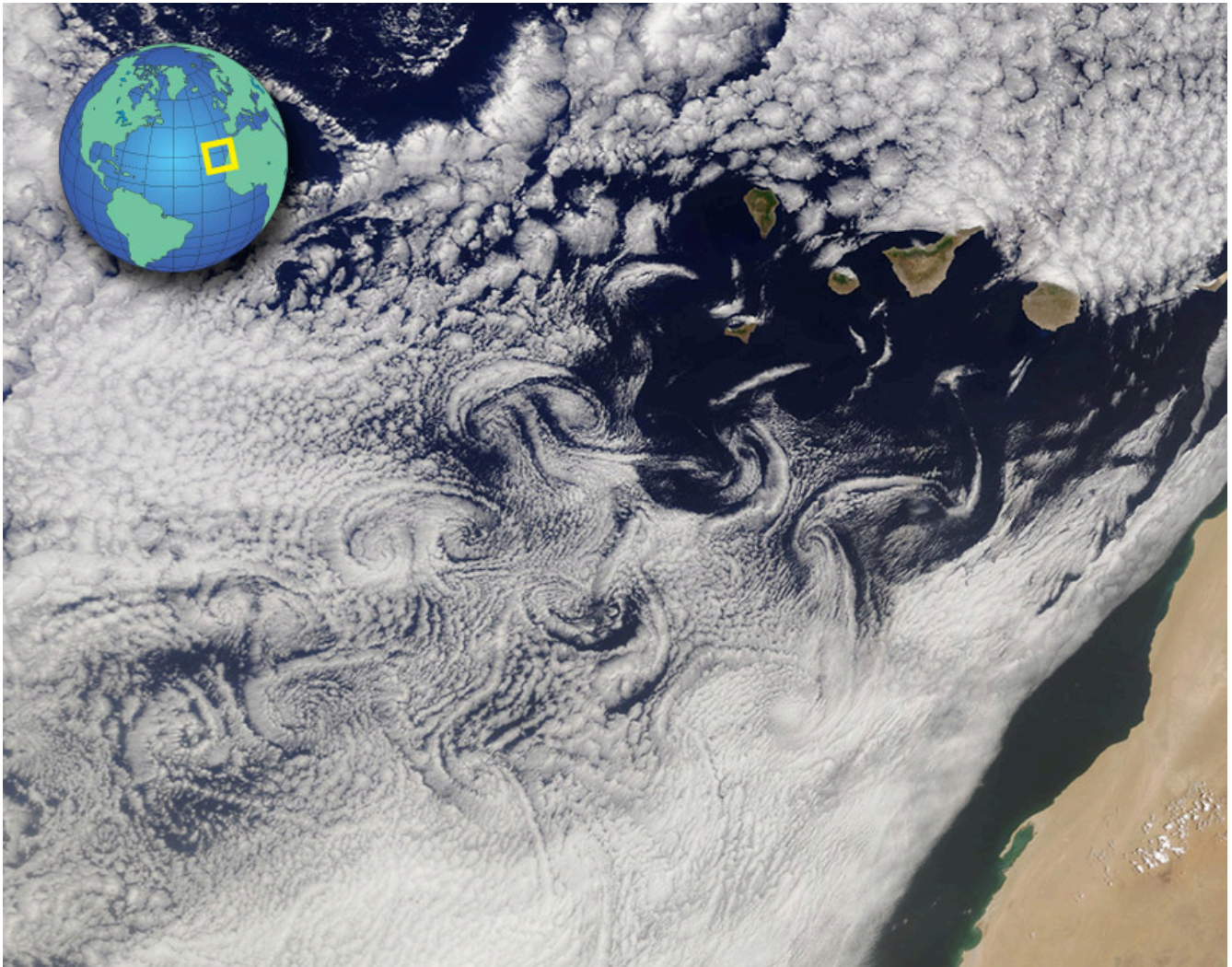


JetStream - An Online School for Weather

Welcome to JetStream, the National Weather Service Online Weather School. This site is designed to help educators, emergency managers, or anyone interested in learning about weather and weather safety.

Questions or comments? Please email JetStream@noaa.gov.

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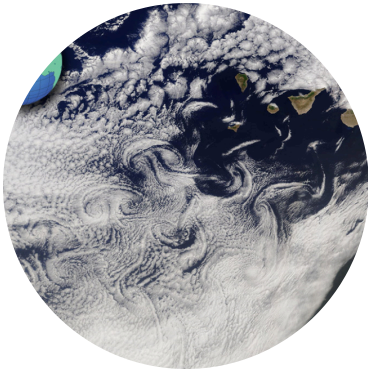
A cloud-filled sky, rugged islands, and turbulent air joined to create fanciful designs in the atmosphere in late July 2019. The Moderate Resolution Imaging Spectroradiometer (MODIS) on board NASA's Terra satellite acquired a true-color image of cloud vortices off of the Canary Islands on July 22.

These particular types of swirling patterns are known as von Kármán vortices. Named after Theodore von Kármán, the Hungarian-American physicist who was the first to describe the physical processes that create the long chains of spiral eddies, these shapes can form nearly anywhere that the flow of fluid or air is disturbed by a stationary object.

In this case, driving wind rushes past the tall peaks on the rugged Canary Islands. As the winds are diverted around the high areas, turbulence is created, and the disturbance in the flow continues downstream in the form of rows of vortices that alternate their direction of rotation. [High resolution version of this NASA image: select link to enlarge image \(~2 mb\)](#). Credit: Jeff Schmaltz, Moderate Resolution Imaging Spectroradiometer (MODIS) Land Rapid Response Team, National Aeronautics and Space Administration (NASA), Goddard Space Flight Center (GSFC).

The information contained in JetStream is arranged by subject, beginning with global and large-scale weather patterns followed by lessons on air masses, wind patterns, cloud formations, thunderstorms, lightning, hail, damaging winds, tornados, tropical storms, cyclones, and flooding. Interspersed in JetStream are "Learning Lessons" which can be used to enhance the educational experience.

You are free to use the materials in any manner you wish. We welcome your feedback on this project. Your input will greatly assist others in teaching the "hows" and "whys" of weather. Not sure where to begin? Click to see all topics in JetStream in the [Topic Matrix](#).



ABOUT JETSTREAM //

The National Weather Service understands the critical value of fast, accurate weather and water information. We know that information is power – the power to save your life or the lives of your loved ones.

JetStream Online School for Weather provides educators, emergency managers, or anyone interested in learning about weather and water with comprehensive, well-organized, colorfully illustrated curricula designed to help teach about the wonders and dangers that abound in the Earth's atmosphere.

[Learn more about JetStream >](#)

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